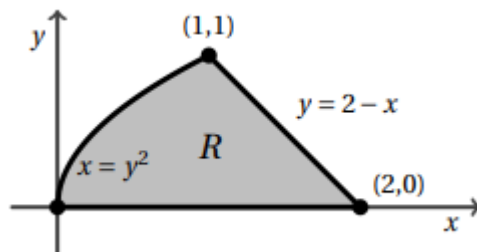


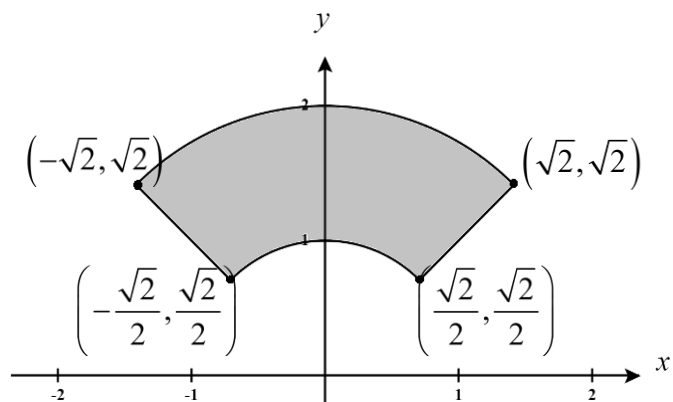
4.8 Centers of Gravity Using Multiple Integrals Worksheet

1. A lamina occupies the region $D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq \sqrt{x}\}$. Find the mass and center of mass of the lamina if its density at the point (x, y) is given by $\rho = 6xy$.

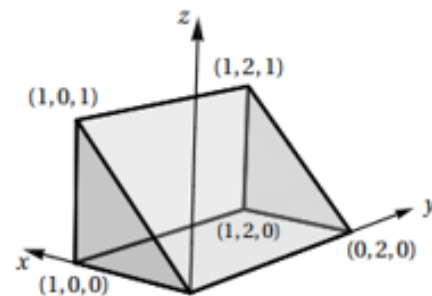
2. Consider the region R pictured to the right. Determine the center of mass of this region if the region has a density determined by $\rho(x, y) = 4y$



3. Consider the region R pictured to the right. Determine the center of mass of this region if the region has a density determined by $\rho(x, y) = x^2 + y^2$



4. Consider the solid region E shown to the right. Suppose the density at any point within E is given by $\rho(x, y, z) = z$. Determine the center of mass of the of E .



5. Let E be the portion of the positive octant which is inside the unit sphere. Determine the center of mass of the E assuming it has a density function $d(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$.

6. Consider the region E shown below which is bounded the xy -plane, the plane $z - y = 0$ and the surface $x^2 + y = 1$. Determine the center of mass of this cone assuming it has a density function $\rho(x, y, z) = xz$.

